

**PROJECT REPORT:
EMPLOYEE ATTRITION
PREDICTION SYSTEM**

CONTENTS

1. Data Understanding Report

1.1 Problem Statement

Business Context

1.2 Objective

1.3 Target Variable

1.4 Input Variables

1.5 Initial Data Checks

Observations

1.6 Data Cleaning Summary

Unique Columns Removed

Null Value Handling

Duplicate Handling

Error Value Handling

Column Segregation

1.7 Outlier Detection & Treatment

Technique Used: IQR Method

Action

1.8 Exploratory Analysis Summary

Univariate Analysis

Bivariate Analysis

Key Observations

2. Feature Engineering Documentation

2.1 Categorical Encoding

Techniques Used

2.2 Feature Transformation

2.3 Feature Selection

Removed

2.4 Final Feature Set

3. Model Development and Evaluation Report

3.1 Dataset Splitting

3.2 Models Implemented

Naive Bayes

Logistic Regression

Decision Tree

Random Forest

3.3 Model Training

3.4 Model Evaluation

Metrics Used (Classification)

3.5 Results Summary

3.6 Model Selection

Reason

4. Insights and Recommendations Report

4.1 Key Insights

4.2 Feature Importance

4.3 Business Recommendations

Reduce Overtime

Improve Job Satisfaction

Salary Optimization

Work-Life Balance Improvement

4.4 Model Usage

4.5 Future Enhancements

5. Conclusion

1. Data Understanding Report

1.1 Problem Statement

The objective of this project is to build a machine learning model that predicts whether an employee will leave the organization or not.

Business Context

Employee attrition is a major concern for organizations as it leads to increased recruitment costs, loss of experienced employees, and reduced productivity. By predicting attrition in advance, companies can take proactive steps to retain valuable employees and improve workforce stability.

1.2 Objective

- To predict employee attrition (Yes/No)
- To identify key factors influencing employee turnover
- To help HR departments take data-driven decisions

1.3 Target Variable

- **Attrition**
 - Yes → Employee will leave
 - No → Employee will stay

1.4 Input Variables

The dataset contains various employee-related features such as:

- Age
- BusinessTravel
- DailyRate
- Department
- DistanceFromHome
- Education
- EducationField

- EnvironmentSatisfaction
- Gender
- JobRole
- JobSatisfaction
- MaritalStatus
- MonthlyIncome
- OverTime
- WorkLifeBalance
- YearsAtCompany
- TotalWorkingYears
- TrainingTimesLastYear

1.5 Initial Data Checks

The following operations were performed:

- Checked dataset shape using `df.shape`
- Verified data types using `df.info()`
- Generated statistical summary using `df.describe()`
- Inspected data using `df.head()`

Observations

- Dataset contains both categorical and numerical variables
- No major missing values found
- Some columns are irrelevant for prediction

1.6 Data Cleaning Summary

Unique Columns Removed

- EmployeeCount

- EmployeeNumber
- Over18
- StandardHours

Null Value Handling

- Dataset had minimal or no missing values
- No major imputation required

Duplicate Handling

- Checked using `df.duplicated()`
- No significant duplicates found

Error Value Handling

- No invalid or unknown values observed

Column Segregation

- Categorical columns separated
- Numerical columns separated

1.7 Outlier Detection & Treatment

Technique Used: IQR Method

- Calculated Q1 and Q3
- $IQR = Q3 - Q1$
- Defined lower and upper bounds

Action

- Removed extreme outliers from numerical features

1.8 Exploratory Analysis Summary

Univariate Analysis

- Distribution of MonthlyIncome analyzed

- Attrition count observed
- Age distribution studied

Bivariate Analysis

- Attrition vs OverTime
- Attrition vs JobSatisfaction
- Attrition vs MonthlyIncome
- Correlation Matrix with heatmap

Key Observations

- Employees working overtime are more likely to leave
- Lower income employees show higher attrition
- Low job satisfaction leads to higher attrition
- Some of the features are the correlate to each other

2. Feature Engineering Documentation

2.1 Categorical Encoding

Techniques Used

- Label Encoding applied to categorical variables

2.2 Feature Transformation

- Data prepared for model training
- No heavy scaling required

2.3 Feature Selection

Removed

- EmployeeCount
- EmployeeNumber
- Over18
- StandardHours

2.4 Final Feature Set

- Input features stored in **X**
- Target variable stored in **y (Attrition)**

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3. Model Development and Evaluation Report

3.1 Dataset Splitting

- 70% Training data
- 30% Testing data

3.2 Models Implemented

1. Naive Bayes
2. Logistic Regression
3. Decision Tree
4. Random Forest

3.3 Model Training

Models were trained using:

- `model.fit(X_train, y_train)`

3.4 Model Evaluation

Metrics Used (Classification)

- Accuracy
- Confusion Matrix

3.5 Results Summary

Model	Accuracy
Naive Bayes	~76%
Logistic Regression	~80%
Decision Tree	~82%
Random Forest	~88%

3.6 Model Selection

Selected Model: Random Forest

Reason

- Highest accuracy
- Better performance
- Handles complex relationships

4. Insights and Recommendations Report

4.1 Key Insights

- Overtime is the most significant factor
- Low salary increases attrition
- Poor work-life balance affects retention
- Job satisfaction is critical

4.2 Feature Importance

Top features influencing attrition:

1. OverTime
2. MonthlyIncome
3. JobSatisfaction

4. WorkLifeBalance

5. YearsAtCompany

4.3 Business Recommendations

1. Reduce Overtime

- Balance workload among employees

2. Improve Job Satisfaction

- Employee engagement programs

3. Salary Optimization

- Provide competitive compensation

4. Work-Life Balance Improvement

- Flexible working policies

4.4 Model Usage

- Input: Employee details
- Output: Attrition prediction (Yes/No)

4.5 Future Enhancements

- Use advanced models (XGBoost)
- Deploy as web application
- Add dashboard for HR

5. Conclusion

The project successfully developed a machine learning model to predict employee attrition. Among all models, Random Forest achieved the best performance. This system can help organizations reduce employee turnover and improve decision-making.